

1 Assignments

1.1 Assignment 1

Problem 1

Question. Consider the following two Monte Carlo schemes for estimating $\mu = \mathbb{E}[h(X) + f(X)]$. The total sample size is $2n$ in both schemes. Scheme 1 (same random numbers): generate $2n$ i.i.d. copies $X_1, \dots, X_{2n}$ and use $\hat{\mu}_1 = \frac{1}{2n} \sum_{i=1}^{2n} (h(X_i) + f(X_i))$. Scheme 2 (different random numbers): write $\mu = \mathbb{E}[h(X)] + \mathbb{E}[f(X)]$, generate $X_1, \dots, X_n, Y_1, \dots, Y_n$ i.i.d., and use $\hat{\mu}_2 = \frac{1}{n} \sum_{i=1}^n h(X_i) + \frac{1}{n} \sum_{i=1}^n f(Y_i)$. Show that both estimators are unbiased and $\text{var}(\hat{\mu}_1) - \text{var}(\hat{\mu}_2) = -\frac{1}{2n} \text{var}[h(X) - f(X)]$.

Hence deduce that $\text{var}(\hat{\mu}_1) \leq \text{var}(\hat{\mu}_2)$.

Solution.

$\mathbb{E}[\hat{\mu}_1] = \frac{1}{2n} \sum_{i=1}^{2n} (\mathbb{E}[h(X_i)] + \mathbb{E}[f(X_i)]) = \mathbb{E}[h(X)] + \mathbb{E}[f(X)] = \mu$,
 $\mathbb{E}[\hat{\mu}_2] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[h(X_i)] + \frac{1}{n} \sum_{i=1}^n \mathbb{E}[f(Y_i)] = \mathbb{E}[h(X)] + \mathbb{E}[f(X)] = \mu$.
 Thus both are unbiased. Moreover,
 $\text{var}(\hat{\mu}_1) = \frac{1}{4n^2} \sum_{i=1}^{2n} \text{var}[h(X_i) + f(X_i)]$
 $= \frac{1}{2n} (\text{var}[h(X)] + 2\text{cov}[h(X), f(X)] + \text{var}[f(X)])$,
 whereas independence of the two samples gives
 $\text{var}(\hat{\mu}_2) = \frac{1}{n} \text{var}[h(X)] + \frac{1}{n} \text{var}[f(X)]$.

Therefore $\text{var}(\hat{\mu}_1) - \text{var}(\hat{\mu}_2) = \frac{1}{2n} (-\text{var}[h(X)] + 2\text{cov}[h(X), f(X)] - \text{var}[f(X)])$
 $= -\frac{1}{2n} \text{var}[h(X) - f(X)] \leq 0$.

Hence $\text{var}(\hat{\mu}_1) \leq \text{var}(\hat{\mu}_2)$.

Problem 2

Question. Given that $S(t)$ is a geometric Brownian motion satisfying

$$\frac{dS(t)}{S(t)} = rdt + \sigma dW(t), \quad S(0) = S,$$

where $W(t)$ is a standard Brownian motion under the risk-neutral measure $\mathbb{Q}$:

(a) Let $Y(t) = S^a(t)$ and find its SDE. (b) using that SDE, suggest two Monte Carlo methods to estimate

$$\mathbb{E}[e^{-rt} \max \left(\frac{1}{2} \sum_{i=1}^n Y(t_i) - \frac{1}{2} K, 0 \right)], \quad 0 < t_1 < \dots < t_m < T.$$

One method must give a biased estimator and the other an unbiased estimator.

Solution. (a) By Itô's lemma,

$$\begin{aligned} dY(t) &= \frac{\partial Y}{\partial t} dt + \frac{\partial Y}{\partial S} dS(t) + \frac{1}{2} \frac{\partial^2 Y}{\partial S^2} (dS(t))^2 \\ &= \alpha S^{\alpha-1}(t) dS(t) + \frac{1}{2} \alpha(\alpha-1) S^{\alpha-2}(t) (dS(t))^2 \\ &= \alpha S^{\alpha-1}(t) [rdt + \sigma dW(t)] + \frac{1}{2} \alpha(\alpha-1) S^{\alpha-2}(t) [r dt + \sigma dW(t)]^2 \\ &= S^{\alpha}(t) \left[\left(ar + \frac{1}{2} \alpha(\alpha-1) \sigma^2 \right) dt + \alpha \sigma dW(t) \right] \\ &= Y(t) \left[\left(ar + \frac{1}{2} \alpha(\alpha-1) \sigma^2 \right) dt + \alpha \sigma dW(t) \right], \text{ with } Y(0) = S^{\alpha}. \end{aligned}$$

(b) By Monte Carlo, $V_n = \frac{1}{n} \sum_{i=1}^n V_k$,

where $V_k = \max \left(\frac{1}{n} \sum_{i=1}^m Y^k(t_i) - K, 0 \right)$

and $\{Y^k(t_1), \dots, Y^k(t_m)\}$ is the k th path of $Y(t)$.

Method 1 (Biased Estimator). Discretizing the SDE gives

$$\begin{aligned} Y(t_i) - Y(t_{i-1}) &\approx \left(ar + \frac{1}{2} \alpha(\alpha-1) \sigma^2 \right) Y(t_{i-1})(t_i - t_{i-1}) \\ &\quad + \alpha \sigma Y(t_{i-1}) [W(t_i) - W(t_{i-1})], \\ \text{so } Y(t_i) &\approx Y(t_{i-1}) + \left(ar + \frac{1}{2} \alpha(\alpha-1) \sigma^2 \right) Y(t_{i-1})(t_i - t_{i-1}) \\ &\quad + \alpha \sigma Y(t_{i-1}) \sqrt{t_i - t_{i-1}} Z_i, \quad Z_i \sim N(0, 1). \end{aligned}$$

Thus the k th path is generated by applying this recursion with independent $Z_i^{(k)}$.

Method 2 (Unbiased Estimator). Solving the SDE gives

$$\begin{aligned} Y(t_i) &= Y(t_{i-1}) \exp(A_i + B_i), \\ A_i &= \left(ar + \frac{1}{2} \alpha(\alpha-1) \sigma^2 - \frac{1}{2} \alpha^2 \sigma^2 \right) (t_i - t_{i-1}), \\ B_i &= \alpha \sigma [W(t_i) - W(t_{i-1})], \\ &= Y(t_{i-1}) \exp \left[\left(ar - \frac{1}{2} \alpha \sigma^2 \right) (t_i - t_{i-1}) + \alpha \sigma \sqrt{t_i - t_{i-1}} Z_i \right]. \end{aligned}$$

Thus the k th path is generated from the exact recursion with independent $Z_i^{(k)}$.

Problem 3

Question. Simulate X by the inverse transform algorithm

$X = \text{int}(10U(1-U))$, where $\text{int}(x)$ is the integral part of x and $U \sim \text{Unif}(0, 1)$. Find $P(X = 0)$, $P(X = 1)$ and $P(X = 2)$.

Solution. Since $U \in [0, 1]$, $10U(1-U) \in [0, 2.5]$, so X can take only 0, 1, 2.

$$\begin{aligned} P(X = 0) &= P(10U(1-U) < 1) = P(10U(2-U) > 1) \\ &= P(0 < U < \frac{1}{2} - \sqrt{\frac{60}{100}}) + P\left(\frac{1}{2} + \sqrt{\frac{60}{100}} < U < 1\right) \\ &= \frac{1}{2} - \sqrt{\frac{60}{100}} + 1 - \left(\frac{1}{2} + \sqrt{\frac{60}{100}}\right) = 1 - \sqrt{\frac{3}{5}} \approx 0.2254. \\ P(X = 1) &= P(1 \leq 10U(1-U) < 2) \\ &= P(10U(1-U) < 2) - P(10U(1-U) < 1) \\ &= P(10U(2-10U) > 2) - P(10U(2-10U) > 1) > 0 \\ &= \left(\frac{1}{2} - \sqrt{\frac{20}{100}} + 1 - \frac{1}{2} - \sqrt{\frac{20}{100}}\right) - \left(1 - \sqrt{\frac{3}{5}}\right) \\ &= \sqrt{\frac{3}{5}} - \frac{1}{\sqrt{5}} \approx 0.3274. \end{aligned}$$

$$\begin{aligned} P(X = 2) &= P(2 \leq 10U(1-U)) = P(10U(2-10U) > 2 \leq 0) \\ &= P\left(\frac{1}{2} - \sqrt{\frac{20}{100}} \leq U \leq \frac{1}{2} + \sqrt{\frac{20}{100}}\right) = \frac{\sqrt{20}}{10} = \frac{1}{\sqrt{5}} \approx 0.4472. \end{aligned}$$

Problem 4

Question. Suppose $Y = (X - 3)^2$, where $X \sim \text{Binomial}(5, 0.5)$. Find the pmf of Y .

Solution. The sample space of X is $\{0, 1, 2, 3, 4, 5\}$ and that of Y is $\{0, 1, 4, 9\}$. Therefore

$$\begin{aligned} P(Y = 0) &= P(X = 3) = \binom{5}{3} (0.5)^5 = \frac{10}{32}, \\ P(Y = 1) &= P(X = 2 \text{ or } 4) = \left[\binom{5}{2} + \binom{5}{4} \right] (0.5)^5 = \frac{15}{32}, \\ P(Y = 4) &= P(X = 1 \text{ or } 5) = \left[\binom{5}{1} + \binom{5}{5} \right] (0.5)^5 = \frac{6}{32}, \\ P(Y = 9) &= P(X = 0) = \binom{5}{0} (0.5)^5 = \frac{1}{32}. \end{aligned}$$

Problem 5

Question. Let X have cdf

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.1x, & 0 \leq x < 1, \\ 0.3, & 1 \leq x < 3, \\ 0.2x, & 3 \leq x \leq 5, \\ 1, & x > 5. \end{cases}$$

Determine the simulated value x resulting from a uniform number u .

Solution.	simulated x
$0 \leq u < 0.1$	$u = 0.1x \Rightarrow x = u/0.1$
$0.1 \leq u \leq 0.3$	$x = 1$
$0.3 < u \leq 0.6$	$x = 3$
$0.6 < u \leq 1$	$u = 0.2x \Rightarrow x = u/0.2$

Problem 6

Question. Generate samples from

$$f(x) = \begin{cases} \frac{1}{2} x^2 e^{-x}, & x \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

by acceptance-rejection, using

$$g(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise,} \end{cases} \quad \lambda \in (0, 1).$$

Determine the best λ minimizing the expected number of trial samples needed to generate a sample from f .

Solution. Let $h(x) = \frac{f(x)}{g(x)} = \frac{1}{2\lambda} e^{-(1-\lambda)x}$.

Then $h'(x) = \frac{-(1-\lambda)}{2\lambda} e^{-(1-\lambda)x} [2x - (1-\lambda)]^2$.

Setting $h'(x) = 0$ gives $x[2 - (1-\lambda)] = 0$, so the positive stationary point is $x = 2/(1-\lambda)$. Since $h'(x) > 0$ before and $h'(x) < 0$ after this point,

$$\frac{f(x)}{g(x)} \leq h\left(\frac{2}{1-\lambda}\right) = \frac{2e^{-2}}{\lambda(1-\lambda)^2}.$$

Thus the expected number of trial samples is $2e^{-2}/[\lambda(1-\lambda)^2]$. Define

$$k(\lambda) = \frac{2e^{-2}}{\lambda(1-\lambda)^2}.$$

Then $k'(\lambda) = -\frac{2e^{-2}}{\lambda^2(1-\lambda)^3}$.

Setting $k'(\lambda) = 0$ gives $\lambda = 1/3$. By the first derivative test, this is the minimum, and $k(1/3) = \frac{2e^{-2}}{(1/3)(1-1/3)^2} = 1.827$.

Problem 7

Question. Suppose X has the standard double exponential density $g(x) = \frac{1}{2} e^{-|x|}$, $-\infty < x < \infty$. Show that $|X|$ has the exponential distribution with mean 1.

Solution. Clearly $P(|X| \leq x) = P(X \leq x)$. For $x > 0$,
 $P(|X| \leq x) = P(-x \leq X \leq x) = P(X \leq x) - P(X \leq -x)$
 $= \frac{1}{2} \int_{-\infty}^x e^{x'} dx - \frac{1}{2} \int_0^x e^{-x'} dx = \frac{1}{2} \int_{-\infty}^x e^{x'} dx = 1 - e^{-x}$.

Hence $f_{|X|}(x) = \frac{d}{dx} P(|X| \leq x) = \begin{cases} 0, & x \leq 0, \\ e^{-x}, & x > 0, \end{cases}$

which is exponential with mean 1.

Problem 8

Question. Let Θ be uniform on $[0, 2\pi]$ and S and $\tilde{S}$ be exponential with mean 2. Assume independence. Show that $X = \sqrt{S} \cos \Theta$, $Y = \sqrt{\tilde{S}} \sin \Theta$ are independent standard normal random variables. Hint: use the two-dimensional change-of-variables formula and its Jacobian.

Solution. Let $g(x, y)$ be the joint density of X, Y , and $f(s, \theta)$ that of S, Θ . Independence gives $f(s, \theta) = \frac{1}{4\pi} \left(\frac{1}{2} e^{-s/2} \right) = \frac{1}{8\pi} e^{-s/2}$.

Since $x = \sqrt{s} \cos \theta$ and $y = \sqrt{s} \sin \theta$, $J = \frac{\partial(x, y)}{\partial(s, \theta)} = \frac{\partial x}{\partial s} \frac{\partial y}{\partial \theta} - \frac{\partial x}{\partial \theta} \frac{\partial y}{\partial s}$

$$= \begin{vmatrix} \frac{1}{2\sqrt{s}} \cos \theta & -\sqrt{s} \sin \theta \\ \frac{1}{2\sqrt{s}} \sin \theta & \sqrt{s} \cos \theta \end{vmatrix} = \frac{1}{2} (\cos^2 \theta + \sin^2 \theta) = \frac{1}{2}.$$

Thus $f(s, \theta) = g(x, y)|J|$ and

$$\begin{aligned} g(x, y) &= 2f(s, \theta) = \frac{1}{2\pi} e^{-s/2} = \frac{1}{2\pi} e^{-(x^2+y^2)/2} \\ &= \frac{1}{(2\pi e)^{1/2}} e^{-x^2/2} \cdot \frac{1}{(2\pi e)^{1/2}} e^{-y^2/2}. \end{aligned}$$

Therefore X and Y are independent standard normal random variables.

Problem 9

Question. Given

$$\Sigma = \begin{pmatrix} 3.21 & 1.82 & 2.18 \\ 1.82 & 5.89 & 3.23 \\ 2.18 & 3.23 & 8.12 \end{pmatrix}, \text{ find its Cholesky factor } A.$$

Solution. Let

$$\begin{aligned} A &= \begin{pmatrix} A_{11} & A_{12} & 0 \\ A_{21} & A_{22} & 0 \\ A_{31} & A_{32} & A_{33} \end{pmatrix}, \text{ and impose } AA^T = \Sigma. \text{ Then} \\ A_{11}^2 &= 3.21 \Rightarrow A_{11} = 1.7916, \\ A_{21}A_{11} &= 1.82 \Rightarrow A_{21} = 1.82/1.7916 = 1.0159, \\ A_{31}A_{11} &= 2.18 \Rightarrow A_{31} = 2.18/1.7916 = 1.2168, \\ A_{21}^2 + A_{22}^2 &= 5.89 \Rightarrow A_{22} = \sqrt{5.89 - (1.0159)^2} = 2.2041, \\ A_{31}A_{21} + A_{32}A_{22} &= 3.23 \Rightarrow A_{32} = 0.9046, \\ A_{31}^2 + A_{32}^2 + A_{33}^2 &= 8.12 \Rightarrow A_{33} = 2.4127. \end{aligned}$$

$$\text{Hence } A = \begin{pmatrix} 1.7916 & 0 & 0 \\ 1.0159 & 2.2041 & 0 \\ 1.2168 & 0.9046 & 2.4127 \end{pmatrix}.$$

Problem 10

Question. Let X, Y be arbitrary random variables and define $\text{var}(X|Y) = \mathbb{E}[X^2|Y] - \mathbb{E}[X|Y]^2$.

Show (a) $\text{var}(X|Y) \geq 0$ and (b) $\text{var}(X) = \mathbb{E}[\text{var}(X|Y)] + \text{var}(\mathbb{E}[X|Y])$.

Hints: (i) $\mathbb{E}[\mathbb{E}[f(X)|Y]] = \mathbb{E}[f(X)]$; (ii) $\mathbb{E}[g(\mathbb{E}[h(X)|Y])k(X)|Y] = g(\mathbb{E}[h(X)|Y])\mathbb{E}[k(X)|Y]$.

Solution. (a) From hint (i), $\mathbb{E}[X|Y]^2 = \mathbb{E}[X|Y]\mathbb{E}[X|Y] = \mathbb{E}[X^2|Y] - \text{var}(X|Y)$.

Therefore $\text{var}(X|Y) = \mathbb{E}[X^2|Y] - 2\mathbb{E}[X|Y]^2 + \mathbb{E}[X|Y]^2 = \mathbb{E}[X^2|Y] - 2\mathbb{E}[X|Y]\mathbb{E}[X|Y] + \mathbb{E}[X|Y]^2 = \mathbb{E}[(X - \mathbb{E}[X|Y])^2|Y] \geq 0$.

(b) $\mathbb{E}[\text{var}(X|Y)] = \mathbb{E}[\mathbb{E}[X^2|Y] - \mathbb{E}[X|Y]^2] = \mathbb{E}[X^2] - \mathbb{E}[\mathbb{E}[X|Y]^2]$, while $\text{var}(\mathbb{E}[X|Y]) = \mathbb{E}[\mathbb{E}[X|Y]^2] - \mathbb{E}[\mathbb{E}[X|Y]]^2 = \mathbb{E}[\mathbb{E}[X|Y]^2] - \mathbb{E}[X]^2$.

Adding gives $\mathbb{E}[\text{var}(X|Y)] + \text{var}(\mathbb{E}[X|Y]) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \text{var}(X)$.

1.2 Assignment 2

Problem 1

Question. Let $S(t)$ be the price of a non-dividend-paying asset under the risk-neutral measure $\mathbb{Q}$: $\frac{dS(t)}{S(t)} = rdt + \sigma dW(t)$,

where r is the continuously compounded risk-free interest rate and $W(t)$ is a standard Brownian motion under $\mathbb{Q}$. Derive the price of a T -year Asian option with payoff $\max(S_G - K, 0)$, $S_G = \left(\prod_{i=1}^n S(t_i) \right)^{1/n}$,

over monitoring dates $0 < t_1 < \dots < t_n < T$. Note: if $\tilde{Y} \sim \mathcal{LN}(\mu, \beta^2)$, then $\mathbb{E}[\max(Y - C, 0)] = \mathbb{E}[\tilde{Y}]N(d_1) - CN(d_2)$.

$$d_1 = \frac{\log(\mathbb{E}[Y|C]) + \beta^2/2}{\beta}, \quad d_2 = d_1 - \beta.$$

Solution. The SDE has solution $S(t_i) = S(0) \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) t_i + \sigma W(t_i) \right]$.

Therefore

$$\left(\prod_{i=1}^n S(t_i) \right)^{1/n} = S(0) \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \frac{\sum_{i=1}^n t_i}{n} + \frac{\sigma}{n} \sum_{i=1}^n W(t_i) \right].$$

Let $X = (W(t_1), W(t_2), \dots, W(t_n))^T$.

$$\text{Then } X \sim N(0, \Sigma), \text{ where } \Sigma = \begin{pmatrix} t_1 & t_1 t_2 & \dots & t_1 t_n \\ t_1 t_2 & t_2 & \dots & t_2 t_n \\ \vdots & \vdots & \ddots & \vdots \\ t_1 t_n & t_2 t_n & \dots & t_n \end{pmatrix}.$$

Let $A = (1, 1, \dots, 1)$. Since $\Sigma^{-1}W(t) = AX$, the linear transformation property of the multivariate normal gives $\Sigma_X^{-1}W(t) \sim N(0, A\Sigma A^T)$,

$$\text{where } A\Sigma A^T = \sum_{i=1}^n (2n - 2i + 1)t_i.$$

$$\text{Thus } \left(\prod_{i=1}^n S(t_i) \right)^{1/n} = \exp \left[\log S(0) + \left(r - \frac{1}{2} \sigma^2 \right) \frac{\sum_{i=1}^n t_i}{n} + \alpha Z \right],$$

where $Z \sim N(0, 1)$ and $\alpha^2 = \frac{\sigma^2}{n^2} \sum_{i=1}^n (2n - 2i + 1)t_i$.

$$\text{Hence } S_G \sim \mathcal{LN} \left(\log S(0) + \left(r - \frac{1}{2} \sigma^2 \right) \frac{\sum_{i=1}^n t_i}{n}, \alpha^2 \right),$$

$$\text{and } \mathbb{E}^{\mathbb{Q}}[S_G] = S(0) \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \frac{\sum_{i=1}^n t_i}{n} + \frac{1}{2} \alpha^2 \right].$$

Using the given lognormal result, the option price is

$$V = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max(S_G - K, 0)] = e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_G]N(d_1) - KN(d_2)$$

$$= S(0)e^{-rT} \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \frac{\sum_{i=1}^n t_i}{n} + \frac{1}{2} \alpha^2 \right] N(d_1) - Ke^{-rT} N(d_2),$$

$$\text{where } d_1 = \frac{\log(\mathbb{E}^{\mathbb{Q}}[S_G]/K) + \alpha^2/2}{\alpha},$$

$$= \frac{\log S(0)/K + \left(r - \frac{1}{2} \sigma^2 \right) \frac{\sum_{i=1}^n t_i}{n} + \alpha^2}{\alpha}, \quad d_2 = d_1 - \alpha.$$

Problem 2

Question. The dynamics of S_1, S_2 are $\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dX_i(t)$, $i = 1, 2$,

where X_i are standard Brownian motions and $dX_1(t)dX_2(t) = \rho dt$. Suppose $S_1(0) = 5, S_2(0) = 6, \mu_1 = 0.03, \mu_2 = 0.05, \sigma_1 = 0.3, \sigma_2 = 0.2$, and $\rho = -0.2$. Find $\text{Cov}(S_1(0.5), S_2(0.5))$.

Solution. The exact solutions are

$$S_i(t) = S_i(0) \exp \left[\left(\mu_i - \frac{1}{2} \sigma_i^2 \right) t + \sigma_i X_i(t) \right], \quad i = 1, 2.$$

Therefore $\text{Cov}(S_1(t), S_2(t)) = \mathbb{E}[S_1(t)S_2(t)] - \mathbb{E}[S_1(t)]\mathbb{E}[S_2(t)]$

$$= S_1(0)S_2(0) \exp \left[\left(\mu_1 + \mu_2 - \frac{1}{2} \sigma_1^2 - \frac{1}{2} \sigma_2^2 \right) t \right]$$

$$\times \mathbb{E} \left[e^{\sigma_1 X_1(t) + \sigma_2 X_2(t)} \right] - S_1(0)S_2(0)e^{\mu_1 t + \mu_2 t}.$$

At $t = 0.5$ this becomes

$$30e^{0.0075} \mathbb{E} \left[e^{0.33X_1(0.5) + 0.2X_2(0.5)} \right] - 30e^{0.04},$$

